

Problem Set for 17 March 2025

Definition (Linear maps) Let f be a linear map. We restate the definition of image and pre-image (as mentioned in mathematical analysis, I believe) in formal and symmetric terms:

1. (the image) $f_*U = \{f(u) \mid u \in U\}$, where U is the subspace of the domain (domain: where f starts/maps from/originates);
2. (the inverse image) $f^*V = \{u \mid f(u) \in V\}$, where V is the subspace of the codomain (codomain: where f ends/maps to/terminates).

Exercise The most elementary part of linear algebra is to study the operations $\{f_*, f^*, +, \cap\}$. It is an enjoyable time to observe how these simple operations generate identities.

Prove the following (whenever we write f_*X , it is assumed that the linear space X is a subspace of the domain, and similarly for f^*Y):

1. $U \subset f^*f_*U$, when does equality hold for all U ?
2. $f_*f^*V \subset V$, when does equality hold for all V ?
3. $f_*f^*f_* = f_*$,
4. $f^*f_*f^* = f^*$,
5. $f_*(U + V) = f_*U + f_*V$;
6. $f^*(U \cap V) = f^*U \cap f^*V$;
7. Explain $f_*(U \cap V) \subset f_*U \cap f_*V$;
8. Explain $f^*(U + V) \supset f^*U + f^*V$;
9. **(optional)** if $f_*(U \cap V) = f_*U \cap f_*V$ always holds, then f is injective, does the converse hold?
10. **(optional)** if $f^*(U + V) = f^*U + f^*V$ always holds, then f is surjective, does the converse hold?
11. **(optional)** $f_*((f^*U) \cap V) = U \cap (f_*V)$;
12. **(optional)** $f^*((f_*U) + V) = U + (f^*V)$.

Definition

1. $\mathbb{F}[x] = \{\text{polynomials over } \mathbb{F}\} = \{\mathbb{F}\text{-sequences with finite non-zeros}\}$;
2. $\mathbb{F}[[x]] = \{\text{formal power series over } \mathbb{F}\} = \{\mathbb{F}\text{-sequences}\}$.

Neither polynomials nor formal power series are functions in general:

1. for $\mathbb{F} = \{0, 1, +, \cdot\}$, the polynomials $x^2 + x$ and 0 represent the same functions;
2. for $\mathbb{F} = \mathbb{R}$, the formal power series $f(x) = \sum_{n \geq 1} x^n e^n$ diverges almost everywhere.

Problem This exercise (**without Axiom of Choice**) show that how $\mathbb{F}[x]$ and $\mathbb{F}[[x]]$ differs.

1. Show that $\mathbf{Hom}_{\mathbb{F}}(\mathbb{F}[x], \mathbb{F}) \simeq \mathbb{F}[[x]]$.

Hint: a linear map $\varphi \in \mathbf{Hom}_{\mathbb{F}}(\mathbb{F}[x], \mathbb{F})$ is uniquely determined by the image of basis, that is, $\{\varphi(x^k)\}_{k \geq 1}$.

2. Show that $\mathbf{Hom}_{\mathbb{F}}(\mathbb{F}[[x]], \mathbb{F})$ has a subspace which is isomorphic to $\mathbb{F}[x]$.
3. **(Optional)** What lies in $\mathbf{Hom}_{\mathbb{F}}(\mathbb{F}[[x]], \mathbb{F}) \setminus \mathbb{F}[x]$?
4. **(Optional)** Show that $\mathbb{F}[[x]]$ has a complementary subspace in $\mathbf{Hom}_{\mathbb{F}}(\mathbf{Hom}_{\mathbb{F}}(\mathbb{F}[[x]], \mathbb{F}), \mathbb{F})$.
5. **(Optional)** Show that there are no surjective linear maps $\mathbb{F}[x] \rightarrow \mathbb{F}[[x]]$.

Hint: there are surjective set maps from X to $\mathbf{Subsets}(X)$.